

Supplementary Material

Temporal Referent Integrity: A Controlled Diagnostic of Referential Resolution Timing in Tool-Using Agents

Anonymous Submission

Review note. This document supports the main submission with implementation, protocol, and extended-result details. The main paper contains the complete argument; this document supplies extended results. No result reported here changes the primary estimand. Literal prompts, complete runner interfaces, inventories, raw outputs, and executable reports are in the separately uploaded anonymous Code and Data Supplement rather than on an external webpage.

1 Benchmark and Frozen Protocol

The TRI-v3 primary evaluation contains 160 tasks in 20 frozen language-template clusters. Tasks are balanced across anchored and dynamic references and across flip, stable, remove, invalidate, and name-collision transitions. The transfer evaluation contains 80 tasks over four unseen schemas (projects, expenses, inventory, and cloud deployments). The guarded TRI-v4 extension contains 40 conditional tasks and distinguishes action-validity and selector-match guards. The model-facing SQLite trajectory experiment uses a frozen 40-task subset and records every query, refresh, target proposal, mutation attempt, and final database diff.

The task generator evaluates the selector in S_0 , applies a controlled transition to obtain S_1 , and computes the target from the instruction-conditioned resolution status. Each task-specified resolver maps a state to one entity or $\perp$; the frozen generators construct a unique eligible winner, and unresolved ranking ties are outside the inventory. At the refresh boundary, Preserve induces a bound commitment $B(q(S_0))$, whereas Reevaluate retains an unresolved query $U(q)$ for evaluation in S_1 . This semantic status need not be serialized as an explicit field. Paired items share their schema, entities, transition, and action; only temporal discourse varies. Every controller receives the instruction’s natural-language selector. Gold targets, generator-normalized selector fields, and pre-/post-refresh winner IDs are not passed to the model.

1.1 Evidence chronology and status

The manuscript distinguishes the original primary comparison from later replication and audit analyses. The chronology below prevents post-primary reanalyses from being read as preregistered primary estimands.

Stage	Frozen status	Role in the manuscript
v3 language clusters	Primary/frozen before calls	Package-level Qwen Generic vs. Lifecycle-Gated estimand; GLM replication
v3 component addenda	Post-primary; frozen before their own calls	Free actor, validity gate, mode-only, untyped-plan, and exact-CTA audits
v7 core replication	Post-primary replication; frozen before calls	New schemas and states; tests directional substitution
Matched full-history baselines	Post-primary strong baseline; frozen before own calls	Ordinary and final-step-aware history across three model families
Human construct labels	Post-primary construct validation	Three blind labels per 100 original/rewrite items
Human-rewrite model runs	Post-primary replication; frozen before own calls	Unchanged controllers on 50 volunteer rewrites
40-task model-facing SQLite	Secondary/frozen execution test; GLM later replication	Model-issued mutations and final database states
v7 deterministic replay	Post-primary, zero API	Consequence verification over frozen target outputs, not a new behavioral replication
PairAcc and identifiability	Post-primary, zero API	Matched, marginal, and shared-eligible reanalyses of frozen outputs
Selection regret	Post-primary, zero API	Proxy maximizers and changed-PairAcc regret in frozen candidate sets
Rule v2	Post-hoc	Benchmark-aware deterministic audit after v1 failure inspection
Public-suite audits	Post-primary/descriptive	Coverage checklist for audited ToolSandbox, AppWorld, and τ^3 versions

Table 1: Evidence status used throughout the paper. Planned or unverified analyses are not reported as results.

1.2 Diagnostic matrix and metric roles

The actionable referential core crosses authorization with whether the selector winner changes. This separates structural sensitivity from success caused by an unconditional policy. Remove and Invalidate are retained as a separate execution-policy slice because Preserve may correctly yield REJECT rather than an entity ID.

Authorization	Transition	Gold target	Diagnostic role
Preserve	Stable	$q(S_0) = q(S_1)$	Ordinary-execution control
Reevaluate	Stable	$q(S_1) = q(S_0)$	Detect refresh overreaction
Preserve	Changed winner	$q(S_0)$	Detect temporal rebinding
Reevaluate	Changed winner	$q(S_1)$	Detect unconditional locking
Preserve	Remove/Invalidate	$\perp$ by policy	Validity/fallback compliance
Reevaluate	Remove/Invalidate	$q(S_1)$	Valid reselection control

Table 2: Complementary condition design. “Changed winner” comprises Flip and name collision; paired rows keep both states, selector, action, schema, and update fixed.

Exact target accuracy measures end-to-end task success but can hide complementary errors. For N complete pairs, we therefore report

$$\text{PairAcc} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{y}_i^P = y_i^P \wedge \hat{y}_i^R = y_i^R].$$

Preserve and Reevaluate marginal accuracy expose which side fails; PairAcc requires both members to be correct. API, parse, protocol, and missing-output rows are incorrect under intention to treat. Conditional substitution then isolates a narrower mechanism: among changed-winner Preserve rows, the initial ID must be observably correct and remain present and action-valid before substitution to the distinct refreshed winner

enters the numerator. This numerator counts violations, not successful preservation of TRI. Wrong writes, invalid attempts, and unnecessary rejection are separate execution outcomes, not aliases for target accuracy.

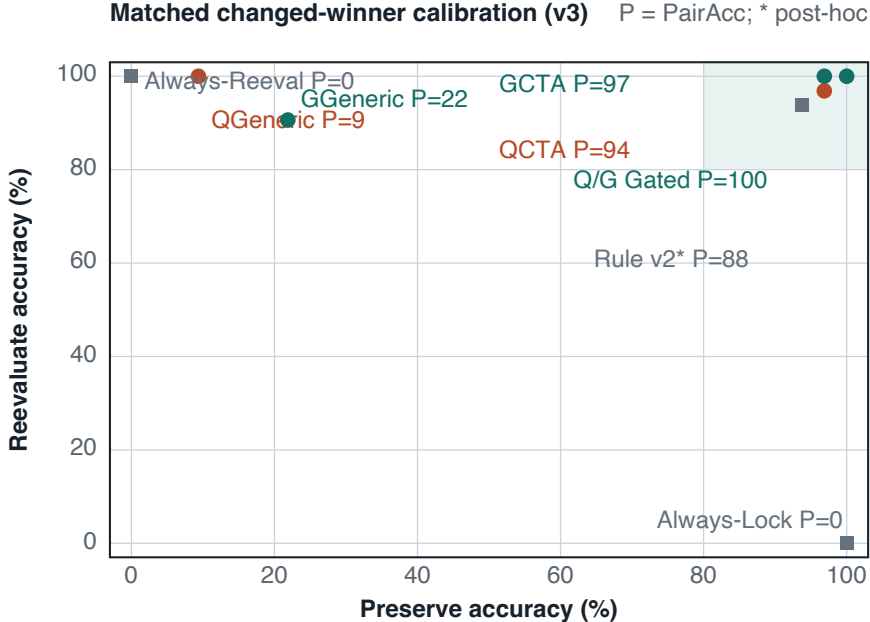


Figure 1: Preserve versus Reevaluate accuracy on the 32 frozen v3 changed-winner pairs. Each model/controller point is labeled with PairAcc (P). The unconditional controls occupy opposite corners, while high joint performance requires the upper-right region. Coincident Qwen/GLM Lifecycle-Gated points are combined. Rule v2 is post-hoc. Values are generated from the same raw runs as the matched-pair tables.

Question	Evidence	Supported conclusion
Stable Preserve/Reevaluate semantics	Blind labels and independent rewrites	Core scalar construct supported; reject fallback weaker
Substitution after correct binding	v3/v7 conditional audits, three models	Occurs for tested controllers under controlled conditions
Refreshed-winner direction	Flip, Stable, and name-collision controls	Supports a controller-level diagnosis, not random error
Executed consequence	Deterministic SQLite replay	Can cause wrong-entity mutation
Dependence on CTA	CTA, Lifecycle, and post-hoc Rule v2	No uniquely necessary implementation
External-style occurrence	Mixed ToolSandbox-compatible audits	Limited and not reproduced at lower intervention
Public benchmark coverage	ToolSandbox, AppWorld, and τ^3 audits	Zero strict native opportunities in audited versions
Natural prevalence	No uncontrolled traffic evidence	Not estimated

Table 3: Evidence boundary. Each row separates the completed evidence from the strongest conclusion it supports.

2 Controller Records

Executable resolution status and transition authorization are the design principle. Historical CTA is the minimal scalar pre-refresh commitment implementation and has no deterministic gate. Lifecycle adds a typed, auditable validity/fallback record; Lifecycle-Gated enforces that record at mutation time. The Generic pipeline

passes the original instruction separately to its actor; its compiled ledger stores the initial selected ID, full entity snapshot, selector, requested action, and action preconditions. It omits reference mode and invalidity/fallback semantics. The scalar Lifecycle compiler additionally emits `reference_mode`, `bound_target_id`, `selector`, and `invalidity_policy`. A deterministic executor checks action validity for preserved identities and delegates only explicitly dynamic or unresolved branches to the actor. The separate TRI-v4 conditional extension adds `guard_type` and `fallback_policy`; these fields are not part of the scalar v3 record. The exact historical Compile-then-act baseline is retained unchanged as a neighboring audit.

The v4 guards are deterministic. Under `action_validity`, the controller retains the old ID iff all action preconditions $V_a(e_0, S_1)$ hold; otherwise it reevaluates $q(S_1)$. Under `selector_match`, it retains the old ID iff it is action-valid and remains $q(S_1)$; otherwise it reevaluates. The frozen 40-task inventory crosses two guard types, five instruction templates per guard, four unseen schemas, and scheduled transition types. Qwen Generic scores 21/40 (52.5%) and Lifecycle-Gated 34/40 (85.0%); this secondary extension is not part of the primary comparison. Complete task rows and the source-derived report are in the Code and Data Supplement.

2.1 Lifecycle-Gated procedure

Lifecycle-Gated execution	
1	Observe S_0 and compile $L(r, a)$ from the instruction and action schema.
2	Execute the required environment refresh and receive S_1 .
3	If mode is PRESERVE, check the compiled bound ID against V_a .
4	If valid, emit the bound ID without invoking the actor.
5	If invalid, apply the compiled reject or reselection fallback.
6	For REEVALUATE or unresolved conditional branches, invoke the actor on $L(r)$ and S_1 .

Table 4: Controller procedure. The instruction supplies the natural-language selector; gold, generator-normalized selector fields, and winner IDs are never provided to the compiler or actor.

2.2 Information and execution audit

Controller	Pre-refresh output	Calls	Gate
Generic ledger	ID, snapshot, selector, action, preconditions	2	no
Untyped plan	free-form plan	2	no
Historical CTA	binding time, ID, selector, reason	2	no
Lifecycle free	mode, ID, selector, policy	2	no
Lifecycle gated	same typed record	1 or 2	yes

Table 5: The gated Preserve branch omits the actor call. Generic retains all task-relevant evidence and is not an information-deletion baseline.

Untyped-plan and Lifecycle-free both compile before refresh, use two calls, share the action schema and actor, and differ in whether the intermediate contract is free-form or typed. Historical CTA is not a pure representation match: its actor omits the action schema and its Preserve rule rejects only missing, rather than present-but-invalid, entities.

2.3 Executor-level oracle check

3 Extended Component Audit

The intervention ladder is interpreted through alternatives it can rule out, not by assuming that every added component is necessary. The principal comparisons are:

Generator-supplied representation	Accuracy	Characteristic failure
Bound ID only	41.9	premature binding
Binding time only	58.1	no anchored identity
Bound name only	58.9	alias/name collision
Latest state	64.6	temporal rebinding
Time + ID	74.0	invalidity/conditional policy
Full lifecycle schema	100.0	–

Table 6: Deterministic execution over 246 generated tasks. This establishes sufficiency for generated semantics only, not model performance, minimality, or natural-language validity.

Comparison	Alternative explanation tested	Result that would weaken the TRI interpretation
Generic vs. full history	Missing task evidence	Full history removes changed-winner substitution
Always-Lock vs. Always-Reevaluate	One fixed update policy suffices	Either extreme succeeds on both pair members
Stable vs. changed winner	Generic refresh instability	Comparable errors without a winner change
Validity gate vs. Generic	Action validity alone identifies the referent	Gate removes valid-target substitutions
Untyped plan vs. typed lifecycle	Any extra pre-refresh computation suffices	Call-matched plan closes the lifecycle gap
CTA/lifecycle vs. Rule v2	A learned or complex method is necessary	Deterministic discourse rule matches them
Target choice vs. SQLite replay	Errors are only representational	Substitutions do not change executed entities

Table 7: Falsification role of completed controls. A failed alternative does not prove a unique internal mechanism; it narrows the controller-level behavioral explanation.

3.1 Handcrafted discourse-rule audit

The rule controller is intentionally non-neural. At prediction time it reads only the instruction, S_0 , S_1 , and action schema. It locates selection and refresh events in the instruction, uses event order to choose Preserve or Reevaluate, resolves the selector in the corresponding state, and applies action preconditions to a preserved ID. It cannot read the benchmark **binding**, **selector**, **update**, **style**, gold target, pre/post target, new leader, CTA record, or any model output.

Version 1 leaves 52/160, 12/50, and 44/240 items unresolved. Version 2 adds the observed event vocabulary and handles a single action-valid candidate without requiring a ranking field. No further changes are made after v2. Its frozen source SHA-256 is `34cbbd39072b...1bbfb8`; the complete digest is in the artifact manifest. On v3, CTA-minus-rule-v2 is +2.5 points for Qwen and +3.8 for GLM, with both cluster intervals spanning zero. On v7, v2 exceeds Qwen CTA and is statistically tied with GLM and DeepSeek CTA. Thus the benchmark does not require a uniquely learned algorithm. The defensible commonality is an executable transition-authorization decision; the rule does not show that a fixed lexicon generalizes to unrestricted instructions.

The post-freeze Generic+reference-mode ablation adds only the explicit mode field and reaches 75.0% for both models. Its paired cluster gains over Generic are +10.6 [+5.0, +16.9] for Qwen and +3.1 [−2.5, +9.4] for GLM; the latter includes zero. The ablation therefore supports explicit mode classification as a component, but not as a complete explanation of CTA’s gain. The audit separates three effects. A validity-only gate changes Generic accuracy by only 0.6 and 1.2 points. Lifecycle state with a free actor reaches 96.9 and 98.1%, while the deterministic gate adds 1.2 and 1.9 points. Exact Compile-then-act reaches 95.0 and 96.2%. The call-matched untyped plan reaches 81.2 and 70.6%; Lifecycle-free exceeds it by 15.6 and 27.5 points with cluster intervals excluding zero. Together with the competitive handcrafted rule, the evidence associates the

Controller	Qwen3.5	GLM-5.1
Generic free actor	64.4	71.9
Generic + reference mode	75.0	75.0
Generic + validity gate	65.0	73.1
Untyped pre-refresh plan	81.2	70.6
Handcrafted rule v2 (post-hoc)	92.5	92.5
Exact Compile-then-act	95.0	96.2
Lifecycle free actor	96.9	98.1
Lifecycle + gate	98.1	100.0

Table 8: Exact target accuracy on the identical 160-task language set.

Rule version	v3 tasks (160)	Human rewrites (50)	v7 tasks (240)
Naive v1, frozen	60.6	74.0	77.5
Strengthened v2, post-hoc	92.5	96.0	91.7

Table 9: Exact authorized-target accuracy (%). Version 1 was frozen before complete-set execution. Version 2 was written after inspecting v1 failures across these inventories and is a strong benchmark-aware post-hoc baseline, not held-out generalization evidence.

gain with operationalizing transition authorization rather than with pre-refresh timing, tuple serialization, or deterministic gating alone. Typed lifecycle state provides an auditable interface; deterministic enforcement contributes a smaller additional gain.

3.2 Matched-pair consistency

For a complete matched pair, Preserve and Reevaluate share S_0, S_1, q, a , action schema, and update. PairAcc is one only when both outcomes are correct. We pair `explicit_anchor` with `implicit_dynamic` and `implicit_anchor` with `explicit_dynamic`. Missing outputs and API, parse, or protocol failures count as incorrect under ITT; none occur in these source runs. Stable pairs are reported as controls, Flip and name collision as the changed-winner core, and remove/invalidate as a separate invalidity-policy slice. The source-derived report is `reports/matched_pair_consistency.json`.

3.3 Evaluation-regime identifiability audit

This post-primary, zero-API reanalysis asks which scoring regimes can identify selective authorization. ‘Preserve-only’ and ‘Reevaluate-only’ retain one binding mode; ‘Stable-only’ never exercises a changed winner; ‘Changed PairAcc’ requires both members of each matched pair. Conditional substitution additionally requires the controller’s observable compiled ID to match the pre-refresh target. Failed or malformed rows are incorrect under ITT. The complete source-derived reports are `reports/v3_identifiability_regimes_v1.md` and `reports/v7_identifiability_regimes_v1.md`; no model calls were added.

On v7, Always-Lock and Always-Reevaluate both score 66.7% overall and 100% on Stable, while both obtain 0/80 changed PairAcc. Generic Reevaluate-only scores are 56.7%, 95.0%, and 100.0% for Qwen/GLM/DeepSeek, but changed PairAcc is 7/80, 15/80, and 17/80. Conditional refreshed-winner substitution is 43/72, 38/80, and 59/79. CTA removes this observed substitution (0/71, 0/70, 0/70) but retains other errors, so the conditional metric is not a general success score. These results support a bounded non-identifiability claim for the controlled diagnostics, not for all public benchmarks or natural traffic.

We then freeze a policy-selection consequence audit over these two reports. Within each dataset/model family, the candidate set contains that model’s reported controllers plus Always-Lock and Always-Reevaluate. For aggregate, Preserve-only, Reevaluate-only, and Stable-only scoring, we enumerate every exact score maximizer and measure regret against the best changed PairAcc in the same candidate set. Fifteen of 20

Policy control	Overall	Anchored	Dynamic
Always-Lock + validity	96/160 (60.0)	80/80 (100.0)	16/80 (20.0)
Always-Reevaluate	96/160 (60.0)	16/80 (20.0)	80/80 (100.0)

Table 10: Deterministic policy extremes on the same frozen inventory. These controls are not LLM runs or reproductions of a named external agent; they expose the complementary errors of unconditional locking and unconditional selector re-evaluation. The machine-readable report is `reports/policy_extreme_controls.md`.

Model	v3 controller	All (80)	Changed (32)	Stable (16)
Qwen3.5	Generic	23 (28.8)	3 (9.4)	16 (100)
GLM-5.1	Generic	38 (47.5)	7 (21.9)	16 (100)
Qwen3.5	Exact CTA	72 (90.0)	30 (93.8)	16 (100)
GLM-5.1	Exact CTA	74 (92.5)	31 (96.9)	16 (100)
Qwen3.5	Lifecycle-free	75 (93.8)	30 (93.8)	16 (100)
GLM-5.1	Lifecycle-free	77 (96.2)	29 (90.6)	16 (100)
Qwen3.5	Lifecycle-gated	77 (96.2)	32 (100)	16 (100)
GLM-5.1	Lifecycle-gated	80 (100)	32 (100)	16 (100)
Independent	Always-Lock + validity	16 (20.0)	0 (0)	16 (100)
Independent	Always-Reevaluate	16 (20.0)	0 (0)	16 (100)
Independent	Rule v2 (post-hoc)	68 (85.0)	28 (87.5)	16 (100)

Table 11: v3 pairs correct (PairAcc %). The 32 invalidity-policy pairs retained in “All” are reported separately in the machine-readable table.

proxy-maximizer sets include a zero-PairAcc extreme; equivalently, all 15 Stable-only or one-sided maximizer sets contain such an extreme. Maximum worst-case regret is 96.9 points. Aggregate E2E is usually aligned with the best PairAcc candidate, but on v3 GLM it selects Lifecycle-free with 6.2 points less PairAcc than CTA. Worst-case tie handling means that the proxy licenses a poor policy, not that users necessarily choose it. The complete post-primary, zero-API table and frozen input hashes are in `reports/evaluation.selection_regret_v1.md`.

3.4 Post-primary parser-binding-executor decomposition

We reuse frozen v3 Lifecycle-free and Lifecycle-Gated outputs to separate three stages without new model calls. The learned condition uses the compiler’s predicted mode and bound ID; oracle interventions replace only the mode, the pre-refresh bound ID, or both. The deterministic gate is then compared with the observed free actor. Reevaluate branches retain the observed actor output, so these cells diagnose component errors rather than estimate an oracle end-to-end agent.

On the primary 160-task inventory, mode accuracy and Preserve bound-ID accuracy are 160/160 and 80/80 for both Qwen and GLM. Consequently, substituting oracle mode or oracle ID does not improve Gated accuracy: Qwen remains 157/160 and GLM 160/160. Lifecycle-free is 155/160 and 157/160, respectively, isolating the small primary free-actor/gate difference after correct compilation. The unseen-schema Qwen transfer is different: mode remains 80/80, but the Preserve bound ID is only 21/40. Gated accuracy is 66/80 with learned binding and 78/80 when only the bound ID is replaced by the oracle; replacing mode alone leaves it at 66/80. Thus the transfer loss is mainly selector grounding at initial binding, not TRI or mode parsing. Complete source-derived cells are in `reports/v3_oracle_qwen_primary.json`, `v3_oracle_glm_primary.json`, and `v3_oracle_qwen_unseen.json`. This analysis is post-primary and uses oracle fields only for diagnosis, not as a deployable baseline.

3.5 Referential-core versus reject-policy sensitivity

The pre-specified aggregate includes 32 anchored remove/invalidate items whose gold execution outcome is the author-specified `INVALID_BOUND_ENTITY`. Table 13 separates these from the 128 actionable referential-core items without redefining benchmark gold or replacing the aggregate estimand. On the actionable core, Gated-minus-Generic differences are +23.4 points with a template-cluster interval of [+11.3,+38.4] for Qwen and +27.3 [+16.9,+39.8] for GLM. Gated-minus-Historical-CTA is −0.8 [−4.9,+3.1] and +0.8 [0.0,+2.6],

Model	v7 controller	All (120)	Changed (80)	Stable (40)
Qwen3.5	Generic	34 (28.3)	7 (8.8)	27 (67.5)
GLM-5.1	Generic	51 (42.5)	15 (18.8)	36 (90.0)
DeepSeek-V4-Pro	Generic	57 (47.5)	17 (21.2)	40 (100)
Qwen3.5	Exact CTA	59 (49.2)	31 (38.8)	28 (70.0)
GLM-5.1	Exact CTA	106 (88.3)	66 (82.5)	40 (100)
DeepSeek-V4-Pro	Exact CTA	103 (85.8)	64 (80.0)	39 (97.5)
Qwen3.5	Lifecycle-gated	67 (55.8)	40 (50.0)	27 (67.5)
GLM-5.1	Lifecycle-gated	113 (94.2)	73 (91.2)	40 (100)
Independent	Always-Lock + validity	40 (33.3)	0 (0)	40 (100)
Independent	Always-Reevaluate	40 (33.3)	0 (0)	40 (100)
Independent	Rule v2 (post-hoc)	100 (83.3)	60 (75.0)	40 (100)

Table 12: v7 pairs correct (PairAcc %). Lifecycle-gated was not run for DeepSeek, so no value is imputed. Low Qwen pair consistency limits any claim of universal CTA sufficiency.

Run	Actionable core	Reject policy
Qwen Generic	95/128 (74.2)	8/32 (25.0)
Qwen Generic + validity gate	95/128 (74.2)	9/32 (28.1)
Qwen Untyped plan	106/128 (82.8)	24/32 (75.0)
Qwen Historical CTA	126/128 (98.4)	26/32 (81.2)
Qwen Lifecycle-free	123/128 (96.1)	32/32 (100.0)
Qwen Lifecycle-Gated	125/128 (97.7)	32/32 (100.0)
GLM Generic	93/128 (72.7)	22/32 (68.8)
GLM Generic + validity gate	93/128 (72.7)	24/32 (75.0)
GLM Untyped plan	107/128 (83.6)	6/32 (18.8)
GLM Historical CTA	127/128 (99.2)	27/32 (84.4)
GLM Lifecycle-free	125/128 (97.7)	32/32 (100.0)
GLM Lifecycle-Gated	128/128 (100.0)	32/32 (100.0)

Table 13: Correct/total and accuracy in parentheses. The policy slice measures compliance with the benchmark’s explicit execution rule, not equally human-supported referential semantics.

respectively; the cross-model evidence therefore does not establish Gated superiority over the strong pre-refresh predecessor. Complete reject-policy contrasts are in the artifact report.

3.6 Frozen error taxonomy

Error label	Operational meaning
Temporal rebinding	A committed pre-refresh target is replaced by the refreshed selector winner.
Premature binding	A dynamic reference incorrectly retains a pre-refresh entity.
Invalid processed	A preserved entity violates action preconditions but is still selected.
Unneeded rejection	The controller returns $\perp$ although a valid gold target exists.
Selector grounding	Mode or guard is correct, but the compiler or actor misapplies a filtered selector.

Table 14: Mechanism-oriented labels used in stage and write-consequence reports.

4 Statistical and Audit Details

The primary comparison is Lifecycle-Gated minus Generic Structured Ledger on Qwen’s 20 language clusters. We resample complete clusters 10,000 times with seed 20260717. The primary Qwen difference is +33.8 points with a cluster interval of [+18.1,+50.0]; the GLM replication difference is +28.1 points with [+18.1,+38.1]. Exact paired McNemar tests are secondary: Qwen has 3 Generic-only versus 57 Lifecycle-only wins, $p = 6.25 \times 10^{-14}$; GLM has 0 versus 45, $p = 5.68 \times 10^{-14}$. API errors and missing rows are counted as failures in the audit, although the reported v3/v4 complete runs contain zero API errors and zero retries.

4.1 TRI-v7 Core Replication

The post-primary replication was frozen before its model calls with SHA-256 prefix `2504f4979f1b`; the artifact manifest records the complete hash. It contains 240 tasks over ten schemas unused in v3 primary or transfer, four independently parameterized state instances per schema, and balanced binding, update, and explicitness factors. All old targets remain present and action-valid. We resample the 40 state-instance clusters.

Model	Controller	E2E target	Initial bind	Cond. subst.	Wrong writes	Stable err.
Qwen3.5	Generic	47.5	107/120	43/72	44/240	4/40
Qwen3.5	Exact CTA	70.8	103/120	0/71	8/240	6/40
Qwen3.5	Gated	71.2	95/120	0/64	17/240	9/40
GLM-5.1	Generic	70.0	120/120	38/80	38/240	2/40
GLM-5.1	Exact CTA	94.2	104/120	0/70	14/240	0/40
GLM-5.1	Gated	97.1	119/120	0/79	7/240	0/40
DeepSeek-V4-Pro	Generic	73.8	119/120	59/79	60/240	0/40
DeepSeek-V4-Pro	Exact CTA	91.2	107/120	0/70	17/240	1/40

Table 15: Frozen new-state replication with distinct denominators. E2E target uses all 240 rows; initial binding uses 120 anchored rows; conditional substitution requires a correct binding, a changed winner, and a still-valid old entity; wrong writes count all non-gold mutations; stable error uses 40 anchored Stable controls.

Generic conditional substitution is 59.7% for Qwen, state-cluster 95% CI [46.5,72.4], and 47.5% for GLM [35.0,61.3]. CTA-minus-Generic overall differences are +23.3 points [16.2,30.4] and +24.2 [19.6,28.7]. SQLite replay turns all 43 and 38 Generic core substitutions into wrong-entity writes. CTA/Gated produce zero core substitution writes after a correct binding, but still produce 8 and 17 Qwen wrong writes, respectively, and 14 and 7 GLM wrong writes, respectively, from opposite-mode or selector errors. Thus the replication supports the conditional phenomenon and commitment compilation, not general write safety. The original six Qwen/GLM 240-row runs contain zero API, parse, or internal errors. GLM CTA records one successful transport retry; the other five runs record none.

The later DeepSeek run uses the same frozen 240 tasks and inference settings. Generic substitutes on 59/79 opportunities, 74.7% with state-cluster interval [62.5,86.1], while CTA substitutes on 0/70. CTA-minus-Generic accuracy is +17.5 points [10.8,23.3]. All 59 Generic core substitutions become wrong-entity SQLite writes. Across all errors, Generic and CTA produce 60 and 17 wrong writes; CTA’s remaining errors are outside the conditional-substitution definition. Both 240-row files have zero API/parse errors and zero retries.

Model	Shared subst. G/C	Generic PairAcc	CTA PairAcc	CTA−G
Qwen3.5	41/66; 0/66	7/80 [2.5,16.2]	31/80 [26.2,51.2]	+30.0 [16.2,43.8]
GLM-5.1	30/70; 0/70	15/80 [8.8,30.0]	66/80 [73.8,90.0]	+63.7 [52.5,75.0]
DeepSeek	50/69; 0/69	17/80 [11.2,32.5]	64/80 [70.0,88.8]	+58.8 [43.8,72.5]

Table 16: Post-primary zero-API v7 audit. Shared substitution uses only tasks where Generic and CTA both expose the correct initial ID; G/C denotes Generic/CTA. PairAcc entries give counts and state-cluster bootstrap 95% intervals in percent; the final column is the paired percentage-point difference. Zero observed CTA substitutions do not establish zero population risk.

4.2 Matched Full-History Baselines

After the v7 and DeepSeek results were known, we froze a separate strong-baseline protocol on the unchanged 240 tasks. `interactive` is an ordinary two-call trajectory: the first response requests refresh and the second retains the complete conversation, without TRI or reference-mode terminology. `full_history_once` is a strong one-call late-authorization baseline that sees both states and is explicitly asked whether to preserve,

reevaluate, or reject. Exact CTA uses the existing matched v7 run. Temperature is zero, thinking is disabled, and the output cap is 1,200.

Model	Interactive	Aware	CTA	Int. subst.	Aware subst.	CTA-Aware CI
Qwen3.5	63.3	69.6	70.8	77/80	56/80	+1.2 [-6.7,9.2]
GLM-5.1	67.1	80.8	94.2	74/80	38/80	+13.3 [8.8,17.9]
DeepSeek	68.8	75.8	91.2	68/80	42/80	+15.4 [9.2,21.7]

Table 17: Matched v7 exact-target accuracy (%) and anchored changed-winner substitutions. Intervals resample 40 state clusters. For Qwen, the CTA-Aware interval includes zero.

All six files contain 240 rows and zero final API/parse failures. Qwen aware uses 269 request attempts with 29 retries; the other one-call aware runs use 240 attempts and zero retries. Interactive runs use 480 attempts. Since neither full-history condition exposes a separately scored pre-refresh selection, its substitution rate is unconditional and is not a conditional substitution estimate. SQLite replay converts the interactive errors into 87/79/75 wrong writes and the aware errors into 70/46/57 wrong writes for Qwen/GLM/DeepSeek, with four invalid attempts and one unnecessary rejection across all six runs. Thus complete history and a late semantic reminder do not remove wrong-entity consequences consistently across controllers; Qwen’s statistically indistinguishable comparison means the result does not establish universal CTA superiority or pre-refresh necessity.

5 Qualitative Trace Cases

Table 18 places positive, negative, and excluded cases side by side. All rows are extracted from frozen raw outputs and executable state diffs, but were selected after the complete runs for explanation. They demonstrate that the reported behaviors occurred in the tested model-tool loops; they are not a random sample or evidence of natural prevalence.

Case	User instruction (abridged only for layout)	Observable model/tool trace	Classification
v7 / Qwen	Choose the earliest incomplete reminder from the present view; reload; keep that chosen reminder as the action target.	bind REM-1A → target/write REM-1B	Post-binding substitution; controlled SQLite
TS / GLM	Before synchronizing, find the newest-created editable reminder; afterward move that reminder, without switching entries.	bind REM-A → postpone(REM-B)	Post-binding substitution; custom, post-hoc case
AW / GLM	Find the earliest task in the list as it is now; synchronize, then postpone the task found.	bind 6334 → postpone(6334)	Eligible opportunity; no violation
AW / Qwen	Find the alphabetically first note as the notes are now; synchronize, then append to the note found.	sync → late bind/write 3086	Tool-order error; not TRI

Table 18: Source-validated qualitative cases. Exact instructions, raw model actions, full traces, and source paths are in `reports/qualitative_trace_cases.json`. The external rows use custom tasks on ToolSandbox-compatible (TS) or AppWorld-backed (AW) environments and are not official benchmark results.

Frozen v7/SQLite conditional substitution. Task `tri-v7-core-reminders-s1-explicit_anchor-name_collision` asks Qwen3.5 Generic to choose the earliest incomplete reminder now, reload, and postpone that same reminder. In S_0 , REM-1A (display “Renewal,” due 110) wins; the compiled ledger records `selected_entity_id=REM-1A`. In S_1 , REM-1A remains present, incomplete, and actionable, while REM-1B changes to due 103 and the same display “Renewal.” The actor’s final response is `"target_id":"REM-1B"`. Replay executes `mutate_entity(target=REM-1B)` and records `wrong_entity_write`; CTA on the same task emits and writes REM-1A. There are no API, parse, or protocol errors. Sources are `runs/v7_qwen_generic_structured_ledger_then_act_full.jsonl`, `runs/v7_qwen_compile_then_act_full.jsonl`, and `runs/v7_core_sqlite_replay.jsonl`.

ToolSandbox-compatible positive intervention. The post-hoc selected custom scenario ID is

`ts2-newest-created-preserve-flip.`

GLM-5.1 Generic searches [REM-C,REM-A], records `selected_entity_id=REM-A`, synchronizes, then searches [REM-C,REM-A,REM-B]. Although REM-A remains editable, it calls `postpone_reminder(REM-B)` and the database diff records a wrong write. Matched Lifecycle writes REM-A. This uses ToolSandbox’s reminder database and native tools, but it is a custom benchmark-compatible intervention, not an official score. Source: `runs/toolsandbox_tri_matched_glm_heldout_v1.jsonl`.

Opportunity without violation. In custom AppWorld task `appworld-todoist-p1-preserve-flip`, GLM binds task 6334 before synchronization; new task 6336 becomes earliest afterward, but the trace calls `postpone_task(task_id=6334)`. Thus a changed-winner opportunity need not produce a violation. Source: `runs/appworld_tri_Pro_zai-org_GLM-5.1_full_history_full_v1.jsonl`.

Excluded pre-binding write. Qwen task `appworld-simple-note-p1-preserve-flip` first searches, synchronizes, searches again, and only then records refreshed note 3086 before writing it; the intended initial target is 3084. Because no correct pre-refresh binding exists, this is a pre-binding tool-order error, not conditional substitution. Source: `runs/appworld_tri_simple_note_Qwen_Qwen3.5-122B-A10B_full_history_full_v1.jsonl`. The source-derived compact fields and traces for all four cases are in `reports/qualitative_trace_cases.json`.

The Qwen 40-task SQLite run gives Generic Structured Ledger 27/40 final states and 13 wrong-entity writes; the GLM run gives 26/40 and 8 wrong-entity writes. A conditional audit requires a correct initial binding, an anchored flip/name-collision update, continued presence and action-validity of the old entity, and a final write to the refreshed winner. Under this strict definition, core substitution writes occur in 8/8 Qwen and 6/8 GLM opportunities, versus 0/4 stable controls for each model. The remaining 5 and 2 wrong writes follow removal or invalidation and are reported separately as reject-policy errors. On Qwen, Generic plus a validity-only gate remains 27/40, while Lifecycle-free and Lifecycle-Gated reach 40/40. The separately run GLM Lifecycle-Gated replay also reaches 40/40; GLM Lifecycle-free was not run on SQLite. All reported 40/40 cells have zero wrong writes and collateral modifications. These are controlled mutation results, not a claim about arbitrary unconstrained agents: they assume the lifecycle record was compiled correctly.

6 External Boundary Check

The ToolSandbox-based 24-task pilot is exploratory and is not an official benchmark score. For Generic, Lifecycle, Gated, and Untyped, respectively, accuracies are 79.2/87.5/91.7/91.7% for GLM and 91.7/83.3/91.7/79.2% for Qwen. Gated includes one GLM and two Qwen wrong writes. It therefore demonstrates that the target-integrity diagnosis can be instantiated in a stateful external-style tool loop, but does not establish a stable method advantage outside the controlled TRI setting.

A post-hoc strict audit joins each trace to the frozen manifest and counts only Preserve/Flip rows where the compiled initial ID is correct, the old entity remains present, and the refreshed winner is a different stable ID. GLM Generic writes the refreshed winner in 3/6 opportunities, with 0/2 wrong writes on eligible Stable controls; GLM Lifecycle is 0/5. Qwen Generic is 0/6, while Qwen Lifecycle-free is 2/6 and deterministic gate replay reduces this to 0/6. Untyped plans expose no auditable compiled ID and are excluded from this conditional denominator. The violating GLM tasks span newest-created, oldest-created, and alphabetically-last selectors. This is positive evidence that TRI can occur in a benchmark-compatible database/tool substrate, but the inventory has only six selector clusters and the strict audit was specified after the outputs existed. We therefore report task IDs and no confirmatory interval or prevalence claim.

We then ran a separate frozen 96-task single-user-turn ToolSandbox-based extension with a Preserve/Reevaluate $\times$ Stable/Flip design and an auditable initial-binding opportunity. Qwen and GLM full-history agents had zero conditional substitutions in 70 and 73 opportunities; matched Qwen and GLM Generic Ledgers had zero in 64 and 87 opportunities. The runs still produced 6, 13, 5, and 4 wrong-entity writes, respectively, but all were initial selector or tool ordering errors and therefore excluded from the post-binding numerator. This negative external result is retained as a boundary check rather than merged with

the 24-task pilot or interpreted as a prevalence estimate. A separate Qwen-only `generic_state_observed` condition is retained in the artifact as exploratory sensitivity evidence (0/73 substitutions, six wrong writes, and 13 prohibited-schema/process errors). Its unmatched interface and protocol failures exclude it from these four paper-facing conditions; it is not hidden or pooled with them.

6.1 Official opportunity and released-trace audit

We separately audited the unmodified public task inventories rather than multiplying semantic tasks by tool-description augmentations. At ToolSandbox commit `165848b`, 129 semantic scenario families generate 1,032 tool-presentation instances. None contains the strict sequence correct binding, independent external transition, and later substitutable entity mutation. One multiple-user-turn task is TRI-like: after finding all friends and changing those stable person IDs to enemies, “them” refers to the same IDs when changing them back. It has no external refresh, competitor, or selector flip.

The downloaded AppWorld `0.1.3.post1` release contains 732 task instances from 244 generator families. It likewise has no independently scheduled post-binding transition, but Todoist family `8ce6779` supplies a natural action-induced near-match. The Agent selects incomplete tasks assigned to the user, reassigns them, and then must “leave a comment there” on the same task IDs even though reassignment makes them fail the initial selector. We audited all 42 released trajectories for this family across 14 experiment configurations. Among 16 operations on gold task IDs, all 16 subsequent comments preserve the same ID and none substitutes another target. There are 152 omitted gold targets, two assignments to non-gold IDs, and three comments on non-gold IDs; only two trajectories pass every official test. These are upstream selection and coverage failures, not post-binding substitutions after a correct target operation.

At current τ^3 -bench commit `cf71a80`, we further audit 2,449 core tasks: 50 airline, 114 retail, and 2,285 telecom. Airline and retail expose no user-side mutation during conversation. Telecom has 2,250 tasks with user mutation actions, but none carries a customer, line, or bill stable ID, and its single-device/app actions create no competing same-role selector target. The strict native TRI count is therefore zero. Eight telecom tasks are natural dual-control near-matches: the Agent binds an overdue bill, the user pays it, and the Agent resumes an associated line. Bill and line are different referential roles, so these are not TRI. The 26 released result files contain 10,832 trajectories from GPT-4.1, GPT-4.1-mini, Claude 3.7, and o4-mini; 936 trajectory IDs match the payment family, but none supplies a strict TRI denominator.

Strict-opportunity feature	ToolSandbox	AppWorld	τ^3
Stable entity ID	yes	yes	no
Observable prior binding	yes	yes	partial
Independent post-binding transition	no	no	yes
Competing same-role entity	no	partial	no
Changed selector winner	no	partial	no
Old entity remains actionable	yes	yes	no
Later target mutation	yes	yes	no
Evaluable authorized target	yes	yes	partial

Table 19: Coverage checklist for each public benchmark’s closest natural near-match, not all tasks. “Partial” marks a related but non-strict structure. Every cell is justified in `reports/benchmark_coverage_checklist.json`. No suite contains all required features.

The missing conjunction can make these benchmarks unable to measure TRI; it does not show that TRI is frequent in deployed traffic. Conversely, a near-match is evidence of task structure, not of model violation. The source-derived funnel makes attrition explicit: ToolSandbox narrows from 129 families to 43 after entity-mutation filtering, then 26/18/1/0 after transition, prior-selection, stable-ID, and strict-opportunity checks. AppWorld narrows from 244 families to one near-match and zero strict exogenous opportunities; its 42 released trajectories expose 16 auditable post-binding operations and zero substitutions. In τ^3 , 2,250/2,449 tasks contain a user-side mutation but none exposes a stable ID in that mutation. These are sequential author classifications, not an independent recall audit or a marginal feature-prevalence estimate. Full counts are generated in `reports/public_suite_coverage_funnel.v1.json`.

6.2 Custom AppWorld database case study

The custom AppWorld study uses two application/selector clusters. The Todoist cluster was frozen first and uses native `create_task`, `show_tasks`, and `update_task` APIs. A post-primary Simple Note extension uses native `create_note`, `search_notes`, and `add_content_to_note`. Each cluster crosses Preserve/Reevaluate, Stable/Flip, and two paraphrases. Todoist selects the earliest-due incomplete task; Stable adds a later task and Flip an earlier one. Simple Note selects the alphabetically first tagged title; Stable adds **DELTA** and Flip adds **ALPHA**. Old targets remain present and editable. Gold is computed from stable IDs, and evaluators compare post-sync and final database snapshots without an LLM judge.

App	Agent	Strict	Write	Bind	Subst.	Wrong
Todoist	Qwen3.5	6/8	8/8	6/8	0/6	0
Todoist	GLM-5.1	8/8	8/8	8/8	0/8	0
Simple Note	Qwen3.5	6/8	7/8	6/8	0/6	1
Simple Note	GLM-5.1	4/8	8/8	4/8	0/4	0

Table 20: Custom AppWorld boundary study. Across 24 correct, correctly timed bindings, conditional substitution is 0/24. The one Simple Note wrong write follows synchronization before binding and is excluded from that denominator. Explicit sidecar omission causes the remaining strict failures. This is not an AppWorld leaderboard result; each app contributes one selector cluster.

A post-primary lower-intervention addendum removes `record.binding` and all prompt language about TRI, binding, or temporal authorization. Each ordinary selector API returns one stable ID, which is logged automatically as the binding event. Across 32 Qwen/GLM trajectories, strict success is 28/32, gold-target writes are 30/32, and conditional substitution is 0/28 (0/6 in Preserve/Flip). Qwen’s two wrong writes use the same Preserve wording in both apps: it synchronizes before its first selector call, then writes the refreshed winner. Matched Stable rows exhibit the same order defect but happen to write A because the winner is unchanged. This is a reproducible pre-binding temporal-order failure and illustrates why Stable controls and a conditional-substitution denominator are both necessary.

7 Negative Compositional Stress Tests

The frozen Qwen TRI-v5 stress test contains 40 SQLite tasks with two refreshes, a monitoring referent, an unrelated tool call, and a final mutation. Generic obtains 32/40 final states, one wrong write, and four unnecessary rejections. Scalar Lifecycle obtains 28/40, six wrong writes, six unnecessary rejections, and one parse/internal error. The difference is -10.0 points with a template-cluster interval of $[-35.0, 12.5]$. Lifecycle mode accuracy is only 25/40: its scalar record confuses monitoring and action-target roles.

A frozen role-indexed held-out addendum then tests 40 tasks from four unseen schemas. In the matched scalar-versus-role comparison, Qwen improves from 35/40 to 39/40 final states (difference $+10.0$, cluster CI $[+2.5, +20.0]$, McNemar $p = .21875$). GLM intention-to-treat is 40/40 scalar versus 37/40 role-indexed because three role runs fail in transport; a serial recovery sensitivity gives 40/40 for both. Against historical CTA, Qwen obtains 38/40 versus 39/40 final states (33/40 vs. 39/40 target accuracy), and recovered GLM ties at 40/40. These results motivate role indexing but do not support stable superiority over simpler compilation.

8 Post-Primary Method-Upgrade Go/No-Go

After the scalar and compositional results above were known, we froze a 20-task decision set: 16 v7 core tasks over 16 state clusters and four v6 multi-refresh/role tasks. We reused the matching frozen Generic, Exact CTA, Lifecycle, and role-indexed outputs and newly ran an Event Graph compiler (M1) and Event Graph plus Executable Selector (M2) with both models. This is an exploratory method-selection smoke, not a powered confirmatory comparison.

On the same combined tasks, Exact CTA obtains 13/20 for Qwen and 20/20 for GLM. M2 therefore changes accuracy by $+2$ and -2 tasks, respectively, and fails the predeclared requirements that both models

Model	Method	Correct	Schema	Mode	Selector
Qwen3.5	Event Graph	9/20	14/20	12/20	–
Qwen3.5	Executable Selector	15/20	15/20	14/20	15/20
GLM-5.1	Event Graph	20/20	20/20	20/20	–
GLM-5.1	Executable Selector	18/20	18/20	18/20	19/20

Table 21: Post-primary 20-task method-upgrade smoke. Selector reports equivalence on both initial and final states. Parse/schema failures count as incorrect under intention-to-treat.

reach at least 95% schema/selector equivalence and show a consistent direction. We do not promote M1 or M2 to the main method. Exact CTA remains the minimal scalar realization, while the independently frozen v6 evidence above motivates role indexing as a limited compositional extension. The zero-API oracle executor is 440/440 on v3/v7/v6 and blocks all stale/wrong writes in 120 deterministic concurrency sequences; this establishes implementation sufficiency only, not learned compiler performance.

Binding Drift audit. The two projects can be located on two independent decision axes:

Initial binding	Preserve after update	Reevaluate after update
Correct	TRI-Preserve	TRI-Reevaluate
Incorrect	Error propagation/repair	Repair plus authorized reselection

Table 22: Problem map. Binding Drift studies persistence and repair of a step-1 referent; TRI conditions on a correct binding and varies whether later reevaluation is authorized. The lower-right combination is outside the present TRI claim.

We audited the official Binding Drift repository at commit [0e040e0954b1](#). Its documented offline command references two files deleted at that commit; exposing the current `workflows_200.jsonl` under the legacy filename in a temporary directory yields 25/25 assertions with unmodified metric and defense code. Its deterministic re-verification reads a gold target and is an oracle upper bound, whereas Entity Lock corresponds to our Always-Lock extreme. We therefore do not place its oracle score in the learned-method main table. Our information-matched one-shot baseline sees the original instruction and both state epochs and must resolve whether to preserve, reevaluate, or reject; it is a late-authorization TRI baseline, not the official self/cross method. The official LLM-based re-verifiers have not been adapted to expose a separately scored pre-refresh binding or a user-authorized deferred query, so we do not label that comparison as an official reproduction.

After freezing a 20-task symmetric smoke (ten v7 domains, one identical-state Preserve/Reevaluate Flip pair per domain), we ran an explicit author adaptation. The official re-verifier normally receives a short `step1 referent`; using only TRI’s selector would delete the temporal authorization variable, so the official prompt frame receives the complete TRI instruction and refreshed candidates. Gold, mode labels, and CTA state are hidden.

Model	Policy	Overall	Preserve	Reevaluate
Qwen	Lock analogue	10/20	10/10	0/10
Qwen	Self verify	3/20	0/10	3/10
Qwen	Cross (GLM)	10/20	0/10	10/10
Qwen	Frozen CTA	12/20	8/10	4/10
GLM	Lock analogue	10/20	10/10	0/10
GLM	Self verify	10/20	0/10	10/10
GLM	Cross (Qwen)	3/20	0/10	3/10
GLM	Frozen CTA	17/20	7/10	10/10

Table 23: Post-primary Binding Drift author-adaptation smoke. Cross rows reuse the other model’s independent verifier output on the same input, avoiding duplicate calls. This is not an official Binding Drift score. Qwen reverify selects another visible but selector-ineligible entity on 14/20 tasks; GLM selects the refreshed winner on all 20.

Both verifier runs contain 20/20 expected rows, zero API/parse errors, and zero retries (16,088 tokens total). Thus GLM cleanly exposes complementary lock/re-resolution errors, while Qwen mainly exposes an interface grounding failure. CTA is not perfect, but is the only tested policy with nonzero accuracy on both modes for both models. These results do not test initial-misbinding repair and cannot establish superiority on Binding Drift’s original benchmark.

Before extending GLM, we froze the complete v7 inventory, endpoint, prompt, ITT policy, metrics, cluster bootstrap, and interpretation gates. All 240 v7 items are actionable: 120 Preserve and 120 Reevaluate across 40 complete state clusters, with 80 each of Flip, Stable, and name-collision.

Policy	Overall	Preserve	Reevaluate	Pair	Subst.	Prem. lock
Lock analogue	160/240	120/120	40/120	40/120	0	80
GLM reverify adaptation	155/240	39/120	116/120	38/120	80	2
Frozen CTA	226/240	110/120	116/120	106/120	10	4
Rule v2 (post-hoc)	220/240	110/120	110/120	100/120	10	10

Table 24: Complete post-primary v7 author adaptation. Pair requires both Preserve and Reevaluate members under the same state transition to be correct. Subst. counts Preserve substitutions to the refreshed winner; Prem. lock counts Reevaluate retention only when the old target is wrong.

Condition	S_0	S_1	Old selected ID	Transition decision
Reverify adaptation	no	yes	no	implicit from instruction
Full-history aware	yes	yes	recomputed	late LLM decision
Generic ledger	snapshot	yes	yes	implicit in ledger/history
CTA	compiler input	yes	yes	precompiled
Rule v2 (post-hoc)	yes	yes	derived	deterministic event order

Table 25: Information audit. Only the full-history, Generic, CTA, and rule conditions contain enough information to reconstruct the old selector result after a changed-winner transition. This table does not imply that recomputation and an observed binding impose equal computational demands.

GLM reverify completes 240/240 rows with 240 requests, zero retries, zero API/parse errors, and 89,354 provider-reported tokens. The pre-frozen gate labels its observed asymmetry a COMPLEMENTARY POLICY RESULT. A post-run information audit, however, identifies a decisive interface mismatch: the adapted verifier sees the complete instruction and S_1 , but neither S_0 nor a resolved old ID. Binding Drift’s original `step1_referent` is intended to uniquely identify the step-1 target; TRI’s ranking selector does not identify its former winner from S_1 after a Flip. Thus the 39/120 versus 116/120 mode split is expected under the available information and is not a fair CTA performance comparison. We retain the full result as an interface audit, not as evidence of superiority or an official Binding Drift score. The matched full-history and Generic-ledger controls above are the information-matched baselines.

Leave-group-out sensitivity. On the 240-task v7 replication, we recompute the matched CTA-minus-Generic difference after excluding one complete domain or one complete template family. The ten domain exclusions retain ranges of 18.5–25.5, 23.1–25.9, and 15.7–22.2 points for Qwen, GLM, and DeepSeek; the sixteen template-family exclusions retain 20.4–26.3, 20.8–26.1, and 14.2–19.6 points. Every exclusion remains positive. This rules out dominance by one domain or template family, but does not make templated rows independent natural-world observations.

Within-inventory subsampling stability audit. We additionally subsample complete matched clusters without replacement 10,000 times at increasing cluster counts. This is a precision and stability diagnostic, not post-hoc power. On v3, the audit reproduces the full Lifecycle-Gated–Generic point differences of +33.8 points for Qwen and +28.1 for GLM; the pre-specified seed-20260717 intervals are reported in the statistical section above. At only 10 of 20 template clusters (80 tasks), 100% of subsamples retain a positive difference for both models. On v7, the full CTA–Generic differences are +23.3 [16.2,30.4], +24.2 [19.6,28.7], and +17.5 [10.8,23.3] for Qwen, GLM, and DeepSeek. At 10 of 40 state clusters (60 tasks), positive differences occur in 100.0%, 100.0%, and 99.8% of draws; at 20 clusters all three reach 100%. Thus the direction is stable

under cluster subsampling within the observed inventory. This does not increase the number of natural-world opportunities or justify a prevalence claim.

Temperature-zero repeat stability. Before repeat calls, we froze one balanced task from each of the 40 v7 state clusters. The matching subset of the original run is repeat 1; two new complete passes give three measurements for Qwen and GLM Generic and CTA. CTA–Generic remains positive in every model-repeat cell: Qwen differences are +25.0, +12.5, and +22.5 points, while GLM differences are +30.0, +20.0, and +25.0. Generic conditional substitution is 5/10, 5/11, and 7/11 for Qwen and 4/12, 5/12, and 6/12 for GLM; CTA is zero in every denominator. Exact target unanimity across all three passes is 70.0/72.5% for Qwen Generic/CTA and 85.0/97.5% for GLM. Under the frozen rule this is a MIXED result: the method direction and conditional mechanism are stable, but individual outputs are not fully deterministic. The 320 new task-controller executions use 640 requests, zero retries, and 313,692 provider-reported tokens, with zero API/parse errors.

9 Request Cost

On the 160-task primary set, Generic uses 320 requests per model. Lifecycle-Gated uses 80 Preserve compiler calls and 160 Reevaluate compiler/actor calls, totaling 240. Mean client-observed Qwen latency is 2.84 seconds for gated Preserve versus 3.63 for Generic anchored tasks; corresponding GLM means are 3.47 versus 5.18 seconds. On the 40-task SQLite set, Generic uses 80 requests and Gated uses 60. Token usage was not captured and latency is network-dependent, so these are request-efficiency observations rather than a controlled systems benchmark.

For the post-primary DeepSeek v7 run, both Generic and CTA use 480 requests. Provider-reported usage is 264,359 tokens for Generic and 216,446 for CTA; client-observed cumulative latency is 1,651.7 and 1,355.8 seconds. These matched-endpoint observations are reported for provenance, not as a cross-provider systems claim.

10 Blind Human Construct Validation

One volunteer uninvolved in TRI design produced natural-language rewrites of 50 sampled scalar instructions. Three other volunteers independently labeled 100 differently randomized original/rewrite items. Forms used opaque IDs and concealed source task, binding mode, update, variant, and gold. Responses were an entity ID, REJECT, or CLARIFY; confidence was optional. All three files contain exactly the 100 keyed IDs with no missing, duplicate, invalid, or payload-mismatched rows. All four participants were adults and provided informed consent. The collection instructions described voluntary participation and discouraged names or unnecessary personal data. Two participants received modest one-time honoraria and two were unpaid; compensation was arranged individually and was not contingent on response content or agreement with benchmark gold. No sensitive personal data were collected, and the anonymous Code and Data Supplement contains only de-identified responses; private consent and study-administration records are not included.

Slice	n	Maj.-gold	Unanimous	κ	α
All	100	86.0	72.0	.708	.709
Original	50	84.0	70.0	.679	.681
Human rewrite	50	88.0	74.0	.737	.738
Explicit	60	86.7	70.0	.672	.674
Implicit	40	85.0	75.0	.758	.760
Dynamic	50	98.0	96.0	.924	.925
Anchored actionable	30	86.7	63.3	.538	.543
Anchored reject	20	55.0	25.0	.087	.102

Table 26: Human agreement (percent except coefficients). Majority–gold uses all items, treating six three-way ties as not matching gold. Among 94 determinate majorities, accuracy is 91.5%.

Pairwise raw agreement is 79.3%; 4% of items receive at least one CLARIFY. All 72 unanimous items match gold. Annotator-level gold agreement is 87%, 80%, and 89%; confidence was absent for annotator 1 and therefore is not used as an outcome. The rewrite slice does not degrade agreement, and explicit/implicit slices are similar. The main weakness is execution fallback: participants often preserve an invalid entity as the discourse referent while disagreeing whether the action answer should be REJECT. We therefore interpret the study as support for the Preserve/Reevaluate core, not as validation of reject or conditional policy. Complete disagreement rows and the auditable analysis script accompany the Code and Data Supplement.

Human-majority model sensitivity. Of the 50 original items, 46 have a determinate human majority, 42 majorities match benchmark gold, and 35 are unanimously gold-aligned. Re-scoring frozen model outputs against the human majority gives:

Model	Generic	Lifecycle-Gated	Historical CTA
Qwen3.5	32/46 (69.6)	41/46 (89.1)	42/46 (91.3)
GLM-5.1	34/46 (73.9)	42/46 (91.3)	41/46 (89.1)

Table 27: Accuracy against determinate human-majority targets (percent in parentheses).

Gated-minus-Generic template-cluster gaps are +19.6 points with a 95% interval of [+5.0,+36.4] for Qwen and +17.4 [+6.8,+31.6] for GLM (18 clusters, 10,000 draws). Secondary task-level exact McNemar tests give $p = .0117$ and $.0078$. On the 42 majority-gold items, Gated reaches 97.6% and 100%; on the 35 unanimous-gold items it reaches 97.1% and 100%. This sensitivity preserves the Generic-to-compiled gap but does not show Lifecycle-Gated superiority over historical pre-refresh compilation.

11 Independent Human-Rewrite Model Evaluation

After the human study and before model calls, we froze 50 instructions rewritten by one independent volunteer and reran the unchanged four controllers. All eight files contain exactly 50 unique expected IDs, all 400 rows have `status=ok`, and no request retry occurred.

Model	Generic	CTA	Lifecycle-free	Gated
Qwen3.5, benchmark gold	60.0	90.0	88.0	90.0
GLM-5.1, benchmark gold	74.0	98.0	92.0	94.0
Qwen3.5, human majority	60.4	89.6	83.3	85.4
GLM-5.1, human majority	68.8	93.8	83.3	85.4
Qwen3.5, human actionable	68.3	92.7	80.5	82.9
GLM-5.1, human actionable	68.3	92.7	80.5	82.9

Table 28: Accuracy (%) on all 50 rewrite golds, the 48 determinate rewrite majorities, and the 41 majority-actionable items (excluding majority-REJECT).

On the 37 unanimous-gold rewrites, Generic/CTA/Lifecycle-free/Gated reach 75.7/97.3/86.5/89.2% for Qwen and 81.1/100/89.2/91.9% for GLM. On the 25 dynamic items, CTA/Gated reach 96/80% and 100/88%, respectively; complete explicitness, update, template, error-type, and majority-gold slices are in the artifact. CTA-minus-Generic benchmark-gold differences are +30.0 points with a template-cluster interval of [+11.3,+50.0] for Qwen and +24.0 [+8.3,+41.3] for GLM. Gated reaches the same Qwen gold accuracy as CTA but is four points lower for GLM and lower against human majorities for both models. Direct CTA-minus-Gated intervals are 0.0 [−11.1,+11.1] for Qwen and +4.0 [−4.3,+14.8] for GLM; CTA-minus-Lifecycle-free intervals are +2.0 [−8.5,+12.2] and +6.0 [−3.6,+16.7], respectively. None excludes zero, so the experiment does not establish CTA superiority over either typed variant. All five Qwen and three GLM Gated benchmark-gold errors are dynamic items compiled as PRESERVE; one is human-contested. CTA’s errors are predominantly author-specified invalid-target outcomes. The result therefore supports pre-refresh

commitment compilation on this volunteer-authored rewrite set, not Lifecycle-Gated superiority. The execution log discloses two pre-call empty-run failures and the non-balanced health-smoke deviation.

12 Code and Data Supplement Map

The separately uploaded anonymous Code and Data Supplement contains the frozen JSONL task files, raw model outputs, SQLite traces, report scripts, controller implementations, protocol files, SHA-256 manifests, and tests. The directory `tri_artifact/reports/` contains the primary cluster reports, component audits, SQLite replay reports, ToolSandbox pilot report, and claim-provenance records, `claims_to_evidence.csv`, matched-pair and qualitative-case reports, coverage checklists, official opportunity/trace audits, and the AppWorld custom protocol. Running `PYTHONPATH=. python scripts/audit_main_paper_evidence.py` rebuilds the paper-facing evidence audit from frozen raw outputs and source reports and checks the critical diagnostic-table rows and headline claims against the manuscript source. Current submission runners receive credentials through environment variables; credentials are not embedded in or stored with the artifact.

AI-assisted engineering disclosure. Generative AI systems were used for language editing, code assistance, and internal review. They were not treated as authors, citable sources, independent annotators, or independent human evidence. The authors verified the manuscript claims, references, code, and reported results.

Model and environment provenance. The paper-facing API runs use SiliconFlow’s OpenAI-compatible `https://api.siliconflow.cn/v1/chat/completions` endpoint and the exact provider model IDs `Qwen/Qwen3.5-122B-A10B`, `Pro/zai-org/GLM-5.1`, and `deepseek-ai/DeepSeek-V4-Pro`. The provider did not expose immutable weight revisions or serving hardware, which limits exact API replay. Frozen v3/v7 calls use temperature zero, thinking disabled, and a 1,200-token output cap; per-row timestamps, request attempts, retries, latency, and configuration are stored in raw JSONL. Protocol files record timeout, retry, smoke, and stopping rules for each extension, while prompts are literal strings in the versioned runners. Offline reports and tests were verified on macOS 26.5.1 arm64 with Python 3.12.13 and pytest 9.1.1; core generation, scoring, bootstrap, and SQLite replay use the Python standard library. These local details reproduce analysis from frozen outputs, not provider inference.